

# Imperial College London

MENG INDIVIDUAL PROJECT

IMPERIAL COLLEGE LONDON

DEPARTMENT OF COMPUTING

---

## Enhancing Online Examination and Invigilation Dashboards

---

*Author:*  
Kaining Zheng

*Supervisor:*  
Estibaliz Fraca

*Second Marker:*  
Tom Crossland

September 10, 2025

## Abstract

Online examinations have become an integral part of higher education, yet they still present persistent challenges related to academic honesty, grading efficiency, and giving meaningful feedback. This project has enhanced the *AnswerBook* platform developed at Imperial College London, by introducing new post-exam analytics functionalities, exploring automated marking approaches and investigating the design of invigilation dashboards.

A particular focus was placed on analyzing student submission sequences to discover behavioral patterns that are both robust in algorithm and interpretable in pedagogics. For this purpose, two analytical methods were designed and compared. The first method employed transition-vector clustering as a simple and interpretable baseline, while the second combined graph embedding (node2vec) with sequential modeling (LSTM autoencoders) to capture relational structures and temporal dynamics. The evaluation of the real examination dataset showed that although the baseline method cannot produce strongly separable clusters, the graph embedding with sequential modeling approach generated coherent and educationally meaningful clusters. The case study further revealed common exam-taking strategies, such as sequential completion, strategic backtracking, and systematic review.

The contributions of this work are threefold: (1) the development of a data pipeline and visualization tools for extracting and analyzing fine-grained submission logs. (2) the empirical comparison of two sequence analysis methods. and (3) the demonstration of how learning analytics can support instructors by reducing workload, informing exam design and enhancing feedback. Overall, the project shows how advanced analytical techniques can bridge technical innovation and pedagogical practice in computer-assisted assessment.

## Acknowledgements

I would first like to express my deepest gratitude to my supervisors, Estibaliz Fraca and Zaki Amin. Their support has been invaluable throughout this project. At times when I felt uncertain about the research direction, their guidance and thoughtful suggestions helped me to regain clarity and confidence. When experimental results did not turn out as expected, their encouragement motivated me to persist, explore alternative approaches, and ultimately push my work further than I thought possible. Their mentorship has been a source of both intellectual and personal growth, and I am truly grateful for their dedication.

I am also profoundly thankful to my parents, whose unwavering support made it possible for me to pursue my studies abroad. They have given me the opportunity to stand on their shoulders and see the world from a broader perspective. For me, studying overseas has felt like living a second life, a chance that would not have been possible without their sacrifices and encouragement.

My heartfelt thanks also go to my wife, Jingming He. No matter what decisions I made, she has always respected them fully and supported me without hesitation. During my Master's year, she shouldered the immense responsibility of caring for our daughter with patience, love, and strength. Her support has been the foundation that allowed me to devote myself to this project. I owe much of this accomplishment to her, and I am endlessly grateful for the life we are building together.

Lastly, I would like to thank myself for the persistence and resilience that carried me through these years. Success in life is often shaped by chance and opportunity, while failure is the more common companion. What truly sets people apart is not the absence of setbacks but the way they respond to them. Some are defeated by failure, while others retain confidence, persevere, and transform it into a stepping stone. I hope that, whatever challenges lie ahead, I will continue to face them with courage and faith, carrying forward the qualities of determination and resilience that have sustained me so far.

# Contents

|          |                                                                           |           |
|----------|---------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                       | <b>5</b>  |
| 1.1      | Motivation and Background . . . . .                                       | 5         |
| 1.2      | Objectives . . . . .                                                      | 6         |
| 1.3      | Research Questions . . . . .                                              | 6         |
| 1.4      | Challenges . . . . .                                                      | 6         |
| 1.5      | Contributions . . . . .                                                   | 6         |
| <b>2</b> | <b>Background</b>                                                         | <b>8</b>  |
| 2.1      | Learning analytics and EDM . . . . .                                      | 8         |
| 2.2      | Answer sequence analysis and behavioral patterns . . . . .                | 9         |
| 2.3      | Invigilation dashboards . . . . .                                         | 9         |
| 2.4      | Automated marking . . . . .                                               | 10        |
| <b>3</b> | <b>PROJECT ANSWER SEQUENCE ANALYSIS</b>                                   | <b>12</b> |
| 3.1      | System Context . . . . .                                                  | 12        |
| 3.2      | Data collection and Processing . . . . .                                  | 14        |
| 3.3      | Descriptive Statistics and Data Visualizations . . . . .                  | 16        |
| 3.4      | Analysis Methodology . . . . .                                            | 18        |
| 3.4.1    | Method I: Vector-based Clustering . . . . .                               | 18        |
| 3.4.2    | Method II: Graph Embedding with Sequential Modeling . . . . .             | 19        |
| 3.4.3    | Comparing methods . . . . .                                               | 22        |
| 3.5      | Implementation Details . . . . .                                          | 23        |
| 3.5.1    | Development Environment . . . . .                                         | 23        |
| 3.5.2    | Data Processing Pipeline . . . . .                                        | 23        |
| <b>4</b> | <b>Evaluation</b>                                                         | <b>26</b> |
| 4.1      | Evaluation Objectives . . . . .                                           | 26        |
| 4.2      | Experimental Setup . . . . .                                              | 26        |
| 4.3      | Evaluation Metrics . . . . .                                              | 27        |
| 4.4      | Results for Method I: Vector Clustering . . . . .                         | 27        |
| 4.5      | Results for Method II: Graph Embedding with Sequential Modeling . . . . . | 28        |
| 4.6      | Comparative Discussion . . . . .                                          | 32        |
| 4.7      | Summary . . . . .                                                         | 33        |
| <b>5</b> | <b>Discussion: Research Questions Revisited</b>                           | <b>34</b> |
| <b>6</b> | <b>Conclusion and Future Work</b>                                         | <b>36</b> |
| 6.1      | Summary of Contributions . . . . .                                        | 36        |
| 6.2      | Limitations . . . . .                                                     | 37        |

|     |                                                |    |
|-----|------------------------------------------------|----|
| 6.3 | Future Work . . . . .                          | 37 |
| 6.4 | Discussion: Educational Significance . . . . . | 38 |

# List of Figures

|     |                                                                                                   |    |
|-----|---------------------------------------------------------------------------------------------------|----|
| 3.1 | Student submission data flow in Answerbook system . . . . .                                       | 13 |
| 3.2 | data processing pipeline . . . . .                                                                | 16 |
| 3.3 | student submission distribution . . . . .                                                         | 16 |
| 3.4 | submissions per task . . . . .                                                                    | 17 |
| 3.5 | multiple students answer sequence . . . . .                                                       | 18 |
| 3.6 | LSTM pipeline overview . . . . .                                                                  | 21 |
| 3.7 | LSTM sequence processing detailed . . . . .                                                       | 22 |
| 4.1 | elbow plot for Method I . . . . .                                                                 | 28 |
| 4.2 | LSTM optimal k . . . . .                                                                          | 29 |
| 4.3 | LSTM clusters distributions . . . . .                                                             | 30 |
| 4.4 | Two students with highly similar submission paths . . . . .                                       | 30 |
| 4.5 | A cluster of students exhibiting sequential completion followed by<br>systematic review . . . . . | 31 |

# Chapter 1

## Introduction

### 1.1 Motivation and Background

In recent years, online examinations have gradually become an indispensable form of assessment in higher education [KM18, GMD11, HI23]. Especially after the pandemic, their importance has become even more prominent [MRLdlRSSSFC21]. According to the 2021 EDUCASE Horizon Report: Teaching and Learning Edition, higher education may never return to pre-pandemic norms. The report highlights the widespread adoption of hybrid learning models, increased reliance on learning technologies, and the integration of remote teaching and assessment methods as key developments shaping the post-pandemic educational landscape [a2021, Mih23].

While online examination platforms are scalable, flexible and efficient, they also bring some challenges. It is very difficult to maintain academic integrity without personal supervision, and teachers often have difficulty monitoring students in real time. In addition, grading large groups may consume a significant amount of resources, and the lack of post-exam analysis limits the opportunities for feedback and teaching reflection. Therefore, educational institutions are now placing higher demands on online examination platforms, not only to deliver assessments effectively, but also to ensure the integrity, improve the efficiency of marking and provide feedback for teaching.

Answerbook is an online examination platform developed and used by the Department of Computing at Imperial College London. It offers digital exam delivery and marking functionalities. However, the current system lacks sufficient support for real-time invigilation, automated grading and post-exam learning analytics, which are critical for ensuring exam integrity and improving pedagogical outcomes. This project aims to optimize all aspects of the examination process by enhancing the invigilation function, post-exam analysis tools and automatic marking of the Answerbook platform. It improves the user experience for both students and educators and contributes to the field of computer-assisted assessment and learning analysis.

## 1.2 Objectives

The primary objective of this project are as follows:

1. To collect information on the order in which students answer questions during the exam to gain a deeper understanding of student behaviour, cognitive processes and potential indicators of exam integrity
2. To develop post-exam analytics functionalities and visualization of student answers
3. To understand how students interact with Answerbook during the exams and ensure a seamless experience
4. To explore the design considerations for automatic marking, based on a review of existing literature

## 1.3 Research Questions

This project is guided by the following research questions:

1. How can online invigilation dashboards be designed to improve usability, reduce teacher workload and maintain exam integrity?
2. In what ways can student behavioral data (e.g., answering sequence, submission attempts) be visualized and analyzed to provide meaningful pedagogical insights?
3. What are potential design approaches for integrating automatic marking into Answerbook?

## 1.4 Challenges

Several challenges were identified in the scope of this work:

1. Technical challenges: Ensuring scalability, performance, and smooth integration with the existing Answerbook infrastructure
2. Data challenges: Processing and visualizing large amounts of fine-grained exam data while maintaining interpretability
3. Pedagogical challenges: Designing tools that align with instructors' actual needs and teaching practices of teachers
4. Ethical challenges: Addressing issues of student privacy and transparency in the use of invigilation and analytics tools

## 1.5 Contributions

This project makes the following contributions:

1. System enhancement: The design and implementation post-exam analytics tools for the Answerbook platform
2. Empirical insights: Analyze real student exam submission data to identify response behaviors and patterns related to teaching and integrity monitoring



3. Design considerations for automatic marking; Literature exploration of potential methods for integrating automatic marking into the Answerbook platform, which provides a foundation for future development
4. Pedagogical relevance: Demonstrate how data-driven tools can reduce instructors' workload while enhancing the feedback loop between assessment and teaching

# Chapter 2

## Background

To provide a comprehensive understanding of the research context, this literature review chapter is organized to four main areas: 1. Learning Analytics and Educational Data Mining (EDM), which establish the theoretical and technical foundations for analyzing examination data; 2. Answer Sequence Analysis and Behavioral Patterns, which explores how students' interaction data can be used to infer cognitive and engagement states; 3. Invigilation dashboard, which focuses on the design and effectiveness of monitoring tools in the Answerbook platform; 4. Automated Marking, which reviews current approaches for short answer questions. These areas respectively correspond to the key components of this project, including the visualization of the learning process, the analysis of students' behaviors, the optimization of the examination invigilation system and the improvement of marking efficiency, thereby providing theoretical support and practical references for the system design and functional implementation.

### 2.1 Learning analytics and EDM

As we are in the Answerbook system, we can collect a certain amount of data from learners such as each task submission timestamp. These data, when properly processed and analyzed, can yield valuable insights into learner behaviors, knowledge states and educational outcomes.

Educational Data Mining (EDM) is a research field that focuses on developing methods for exploring learners' data to improve learning processes and systems. It encompasses a variety of techniques suited to different applications. The most commonly employed among these include: 1. classification and regression 2. clustering 3. association rule mining 4. outlier detection 5. sequential pattern mining and 6. visualization techniques [BZE17]. These techniques provide the foundation for designing adaptive examination systems like the one explored in this project.

Through interviews with 13 experienced teachers, this study [LJW25] reveals two main patterns when teachers using learning analytic tools: one focuses on monitoring students' participation and performance to promote them to achieve the curriculum goals; Another category focuses on the diverse learning needs and feedback of students, and adjusts the teaching design with the help of data. The research

emphasizes that teachers’ effective use of analytical tools not only relies on technical knowledge, but more crucially lies in the combination of pedagogical knowledge and data insights.

## 2.2 Answer sequence analysis and behavioral patterns

Studies show that students’ behavior-data such as the order of answering questions, time spent per question, and navigation pattern can provide insights to their understanding, engagement, and academic performance. [BZEI17, BS14, RV10]

Answer sequence analysis is a key subfield of Learning Analytics and Educational Data Mining and is located within the broader framework of process mining and sequential pattern analysis in education. It focuses on modeling students’ temporal interaction data such as - the order in which they answer questions, time spent per item, answer changes and navigation behaviors – to gain insights into their cognitive and engagement levels [VTOS24].

In the field of education, particularly in the context of problem solving, the granular actions a student makes-like inputting and then deleting a possible answer- are crucial for understanding their thinking process. The authors believe that introducing human-in-the-loop approach into sequence analysis (SA) can help improve the interpretability of the analysis results. [KSEN<sup>+</sup>22]

## 2.3 Invigilation dashboards

In recent years, the increasing prevalence of online assessments has led to a growing body of research on the design and development of invigilation dashboards which is an interface that aim to support instructors in monitoring student behaviors during examinations. These dashboards typically aggregate multimodal data (such as webcams, gaze direction, tab switch and so on) into visual or analytical summaries. The design challenges lie in balancing real-time monitoring with ensuring usability and safeguarding student privacy while maintaining academic integrity. At the same time, there are significant benefits: effective dashboards can reduce instructors’ workload, facilitate rapid identification of suspicious behaviors and enhance transparency in decision-making. As such, invigilation dashboards represent a key intersection between learning analytics and ethical considerations in educational technology.

The study [NPSC21] conducted a comprehensive review of AI-based proctoring systems (AIPS), summarizing the architectures, design parameters and ethical challenges involved in online invigilation systems. Their findings highlight the tension between academic integrity an user privacy, and suggest hybrid proctoring models that combine automated detection with human oversight. These insights are related to my project, which aims to design the Answerbook proctoring dashboard with a

focus on usability, transparency and reduced teacher workload.

The research [DSY<sup>+</sup>21] created a lightweight browser-based proctoring system that uses webcam-based eye gaze tracking to identify instances of cheating in on-line tests. The system detects anomalous gaze patterns by utilizing the WebGazer library and the One-Class Support Vector Machine (OCSVM) model. In particular, frequent departures from the screen may be a sign of cheating. Its accuracy and F1 score in undergraduate experiments were 89.53% and 92.04%, respectively. This method can be integrated into current web-based exam platforms and doesn't require any additional hardware, despite the fact that it is currently only compatible with Chrome browsers and gaze-based features.

This study [LXWQ21] developed a visual analytical tools to help instructors monitor online exams more efficiently. The system collects webcam video and mouse movement data during exams and it can detect suspicious behaviors (such as abnormal head movements, face disappearance and tab switch actions). Also, The system visualizes these behaviors at three levels: student overview, question-level risk and detailed behavior view, which enables instructors to quickly identify high-risk students.

This paper [GG25] proposes a machine learning based method for detecting in-parallel collusion which is a form of cheating where students collaborate simultaneously during online assessments. The authors enhanced an online quiz platform (iQuiz) to collect more fine-grained clickstream data and extracted seven features (such as question submission time, exam submission time and exam score) to model students from multiple dimensions of behavioral similarity. The experimental results show that using the all feature set and a Random Forest Classifier achieves high detection accuracy (98.8%), which outperforms prior researches that rely on only two or three features. Furthermore, the model has also been indirectly verified in terms of students' seating positions on unlabeled real examination data.

This paper [DSA<sup>+</sup>22] describes a method for detecting cheating in online quizzes and assignments by analyzing detailed student log data such as answering time of each question, answer changes and final scores. This approach combines pairwise timeline analysis and answer similarity comparison to identify suspicious collaboration between students. The authors summarized four common cheating pattern: Full Collaboration, End Check, Answer Key Sharing and Divide and Conquer and provided real-world examples with visualizations to illustrate each pattern.

## 2.4 Automated marking

Automated marking refers to the use of computational techniques to evaluate students' responses, ranging from structured multiple-choice questions to open-ended short answer questions or short passages. As online examinations become more widespread, automated marking systems offer opportunities to improve grading efficiency. However, the design of such systems involves different levels of complexity depending on the question type: while multiple-choice questions are relatively

straightforward, short answers require sophisticated natural language processing and machine learning methods. This section reviews existing approaches to automated marking.

#### 2.4.1 Multiple-Choice Questions (MCQ)

There are not many papers specifically describing the automatic scoring of multiple-choice questions because it is not a particularly complex issue. Therefore, I designed this by my own.

MCQ Auto Marking Engine design:

1. Create a form on the auto marking page so that Instructors can input the answer key for each multiple-choice question. It is optional if they want to input a rubric that some choice could get partial marks.
2. Students' answers will be stored in the PostgreSQL database.
3. Develop an automatic marking of multiple-choice engines Once instructors click the auto marking in the front-end web page, this engine will compare students answer with rubric that was entered by instructors.
4. Show the results of multiple-choice questions score in the front-end page.

Example for one MCQ

```
{
  "task_id": "q1_p1_s1_t2",
  "correct_answers": [B],
  "max_score": 1,
  "rubric": {
    "type": "single",
    "partial_credit": false
  }
}
```

#### 2.4.2 Short Answer Questions

The paper [BGS14] mentioned Automatic Short Answer Grading (ASAG) is an increasingly important task in educational technology. It aims to evaluate students' short natural language responses using computational methods. Burrows et al. conducted a comprehensive survey of this field. They outlined five methodological eras: concept mapping, information extraction, corpus-based methods, machine learning and evaluations. In this paper [BGS14], their review also highlighted six key components in ASAG system pipelines from dataset preparation and NLP preprocessing to model evaluation. This framework offers a solid foundation for positioning subsequent ASAG research, especially in terms of grading strategies and evaluation metrics.

## Chapter 3

# PROJECT ANSWER SEQUENCE ANALYSIS

### 3.1 System Context

Answerbook is a web-based examination system developed at Imperial College London to support large scale online assessments across multiple courses and degree programmes. Its architecture combines a Flask-base backend with PostgreSQL database, and adopts a multi-schema design, with each assessment isolated in its own schema. The back end uses Flask-REST to expose the RESTful APIs, while the front end is primarily implemented using server-side rendering supplemented by JavaScript and styled with the Bootstrap framework. Deployment is containerized using Docker and typically relies on Gunicorn as the WSGI server which allows scalability and easier system maintenance.

Within the Answerbook platform, the central data entity for post-exam analysis is the submission record. This record captures every attempt a student makes while answering assessment tasks. Understanding the lifecycle of submissions is crucial, since later components of this thesis rely on these data for analytics and visualization.

When a student accesses an exam page, the front end JavaScript initializes by loading all task identifiers and retrieving the most recent submission state for each. This can ensure the restoration of partially completed work. Then, the client side attaches listeners to input fields, so that any modification in a task answer automatically triggers a save operation. Before the page is closed, the system invokes an additional safeguard to persist unsaved changes.

The saving mechanism is incremental and event-driven. On each detected change, the front end collects the updated answers and associates them with their task identifier and then sends them to the back end via the submission API. Each request contains a batch of changed tasks rather than the full exam state, thus reducing redundant network and storage overhead.

On the server side, the submission API validates incoming answers against task specifications. Invalid inputs are rejected with error messages, whereas valid answers

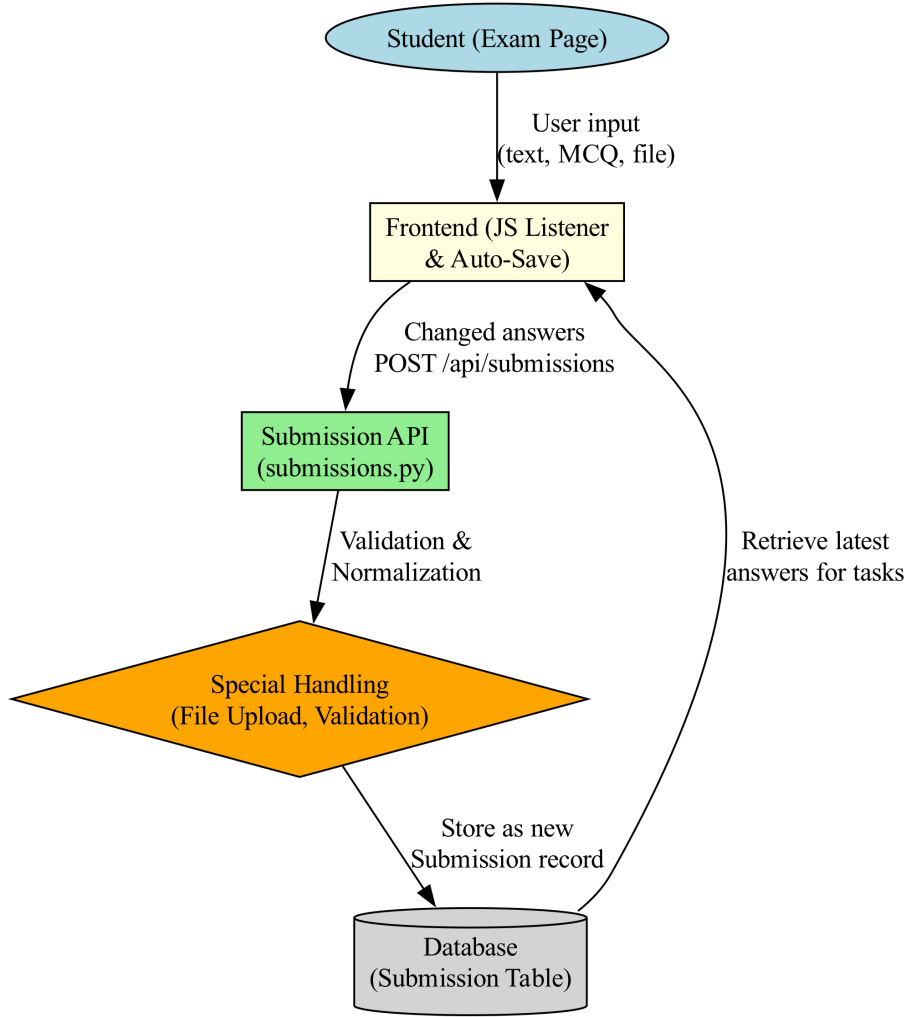


Figure 3.1: Student submission data flow in Answerbook system

are normalized and stored as new submission records. The system adopts an append only storage model. Each change creates a new entry in the database instead of overwriting previous attempts. The database schema reflects that submissions are linking to both the student identity and the exact assessment component (question, part, section and task). Each Submission stores metadata such as timestamp, originating IP address and status flag that indicates whether the submission occurred within the allowed exam time or after closure. The retrieval of answers is always resolved through `Submission.latest()`, which selects the most recent entry for a given task, while still retaining the full submission history for audit and analysis.

This design has several implications: (1) It guarantees that no student work is lost as all intermediate states are preserved. (2) By reconstructing the sequence of attempts, it can conduct a robust post-exam analysis of student behaviour. (3) It simplifies integration with grading and monitoring tools since the final submitted answer can be deterministically retrieved. These properties make the submission subsystem a pivotal foundation for the enhancements explored in this thesis, particularly in the domains of answer sequence analysis and visualization.

## 3.2 Data collection and Processing

The empirical data employed in this study were obtained from the AnswerBook examination platform, which stores exam records in a PostgreSQL database. The analysis focused on the `submission` table, which stores detailed records of students' interactions during the exam. Each record originally contains the student identifier (`username`), a hierarchical specification of the exam item (question, part, section, task), the submission timestamp (`timez`) and the submitted answer content (`answer`). To ensure privacy and confidentiality, all student identifiers were anonymized prior to analysis and the textual content of answers was not retained. Instead, the `answer` field was converted into a simple numerical feature representing the length of the submission. This transformation provided a proxy indicator of engagement while guaranteeing that no personally identifiable information or sensitive exam content was used in the study.

Data was extracted from the database through direct SQL queries using the `psycpg2` library, and subsequently exported into JSON format. Two alternative data representations were considered.

The question-oriented view organizes submissions according to the exam structure. It groups together all students' records under each task. An illustrative fragment is shown below:

```
{
  "question1": {
    "part1": {
      "section1": {
        "task1": [
          {"username": "student_001",
           "timestamp": "2022-05-03T09:15:30Z",
           "answer_length": 150},
          {"username": "student_002",
           "timestamp": "2022-05-03T09:18:45Z",
           "answer_length": 89}
        ]
      }
    }
  }
}
```

This representation facilitates question-level statistics such as average attempts or response patterns, and is thus valuable for assessing item difficulty. However, it fragments each student's activity across tasks, which makes it difficult to reconstruct a continuous trajectory of learning behaviour.

The student-oriented sequence view, by contrast, organizes the data chronologically for each student:

```
{
```



```

"student_001": [
  {"question": 1,
   "part": 1,
   "section": 1,
   "task": 1,
   "timestamp": "2022-05-03T09:15:30Z",
   "answer_length": 150},
  {"question": 1,
   "part": 1,
   "section": 2,
   "task": 1,
   "timestamp": "2022-05-03T09:22:15Z",
   "answer_length": 203}
],
"student_002": [
  {"question": 1,
   "part": 1,
   "section": 1,
   "task": 1,
   "timestamp": "2022-05-03T09:18:45Z",
   "answer_length": 89}
]
}

```

This representation preserves the full temporal sequence of interaction for each student and it enables analyses of pacing, task switching and sequential patterns of behaviour. Its drawback lies in the additional effort required to aggregate information at the question level.

Given the focus of this study on sequence oriented analysis, the student-oriented sequence view was ultimately adopted. It provides the most natural foundation for modelling behavioural trajectories, while still retaining the necessary task level metadata.

Following extraction, the data underwent several preprocessing steps. All timestamps were normalized to ISO 8601 format and converted into Pandas datetime objects. Usernames were anonymized through hashing to protect privacy. To eliminate noise, submissions occurring within ten seconds of a prior submission by the same student were discarded, thereby reducing the influence of accidental or redundant clicks.

The final dataset was stored as a Pandas DataFrame with the following attributes: anonymized student\_id, hierarchical task identifiers (question, part, section, task), a composite task code (task\_id), standardized submission time (timestamp), and answer\_length. On top of these raw variables, derived indicators were computed, including the number of submissions per task, mean submissions across tasks, time spent per task and average task completion time. These transformations ensured that the dataset was both privacy-preserving and analytically suitable for

subsequent modelling.

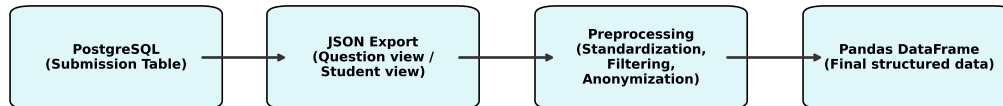


Figure 3.2: data processing pipeline

### 3.3 Descriptive Statistics and Data Visualizations

Before moving to advanced modeling, it is important to provide an overview of the dataset and characterize the foundation properties of student activity. For this purpose, we focus on the 2021 50005 exam dataset, which contains detailed submission records collected from the Answerbook platform.

The dataset includes 158 students, 7791 submissions and 28 tasks. On average, each student made 49.31 submissions, although the distribution was highly skewed: the least active student submitted 10 times, while the most active submitted 133 times. The median submission count per student was 43, and the interquartile range spanned from 28 to 67 submissions. Most students (57.6%) submitted between 21 and 50 times, with 37.3% falling in the 51 - 100 submissions range, which indicates a moderate variation in engagement levels.

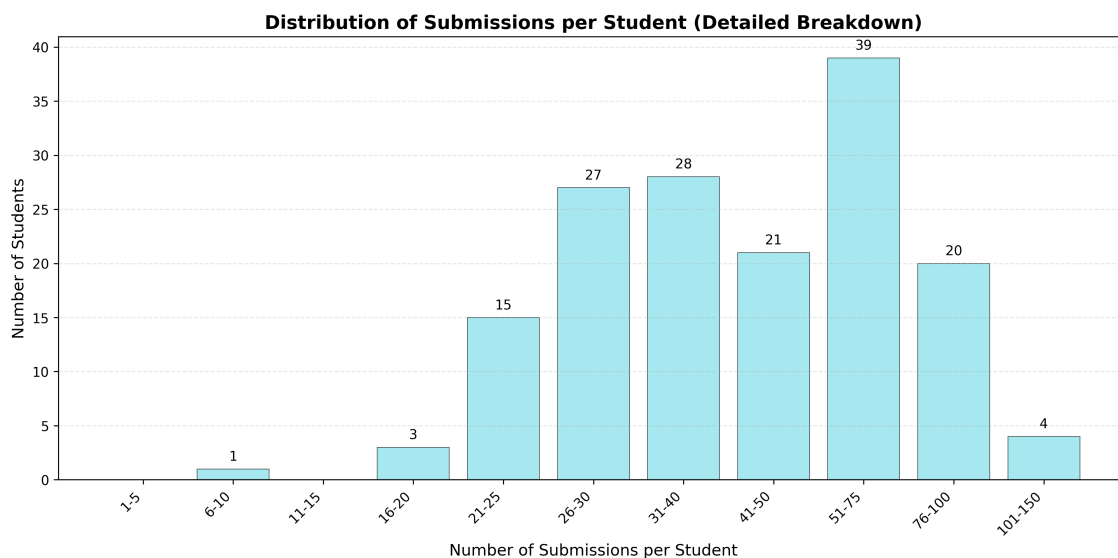


Figure 3.3: student submission distribution

At the task level, submissions were unevenly distributed. On average, each task received 268.66 submissions, with the most popular task receiving 662 submissions

(question 2, part 5, section 1, task 2) and the least popular only 2 (question 2, part 4, section 1, task 1). This suggests significant differences in task difficulty or student engagement across the exam.

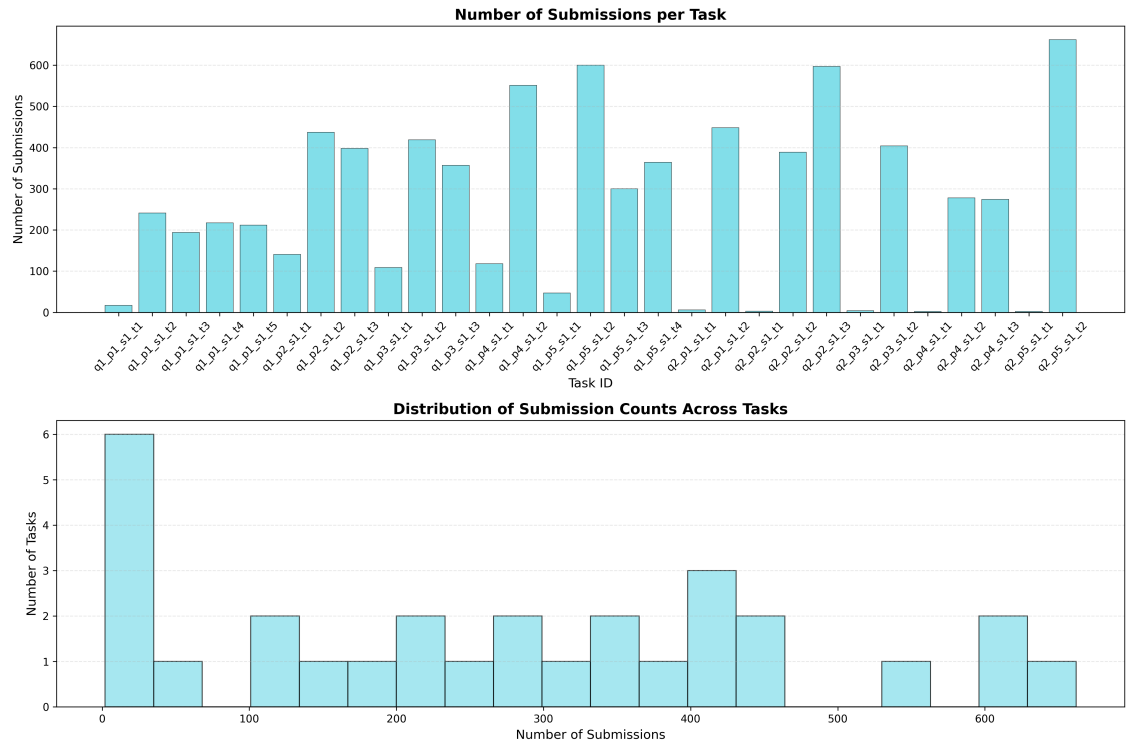


Figure 3.4: submissions per task

The temporal distribution of submissions spanned nearly four hours, from 09:00:43 to 13:00:01, which illustrates the exam’s duration and the pacing of student activity. Such data provide important context for subsequent analyses of sequential behavior and submission patterns.

To better understand student behavior, we visualized submission trajectories as answer sequences using the Plotly library. Figure 3.5 shows an example where each line corresponds to an individual student. The x-axis represents time, and the y-axis represents task identifiers. Interactive features allow hovering over points to reveal metadata such as student ID, task ID and timestamp.

This representation highlights differences in task progression patterns: some students proceed sequentially through tasks, while others jump across sections or revisit earlier questions. Such sequence visualizations provide an intuitive way to detect behavioral diversity, irregular progressions and potential anomalies.

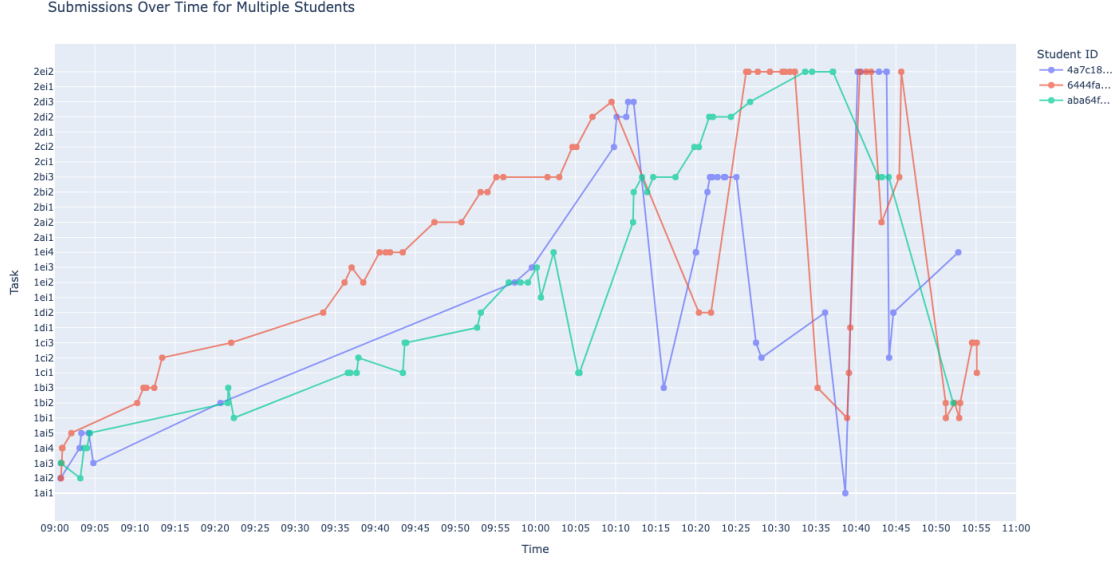


Figure 3.5: multiple students answer sequence

These descriptive statistics and visualizations not only characterize the dataset but also provide the foundation for the more advanced behavioral analyses presented in Chapter 4.

## 3.4 Analysis Methodology

The objective of the analysis is to identify patterns of student submission behaviors, capture the potential structures in sequential task navigation and explore whether clustering techniques can reveal meaningful groups of students that reflect distinct engagement strategies or problem solving approaches. To this end, we have explored three complementary methods, evolving from classical vector-based clustering to deep learning methods with increasing representational power.

### 3.4.1 Method I: Vector-based Clustering

The first method transforms each student’s task submission sequence into transformation matrix, where rows and columns correspond to tasks and entire represent the number of transitions between tasks. This matrix is then flattened into a high-dimensional vector to effectively capture the transition frequencies between tasks.

**Representation.** For each student, let

$$\pi = (t_1, t_2, \dots, t_L)$$

denote the observed task sequence. We form a transition matrix

$$T \in \mathbb{R}^{N \times N}$$

(where  $N$  is the number of tasks) whose entry

$$T_{ij} = \#\{k \mid t_k = i, t_{k+1} = j\}$$

counts transitions from task  $i$  to task  $j$ . We then vectorise  $T$  into

$$x = \text{vec}(T) \in \mathbb{R}^{N^2}.$$

Optionally, rows of  $T$  can be L1-normalised to down-weight prolific students; we report both raw and normalised variants in sensitivity checks.

We run K-means clustering on  $\{x_s\}$  (one vector per student) and use the elbow heuristic on within-cluster sum of squares (inertia) to choose  $k$ . This representation captures coarse navigation behaviour (e.g., “mostly sequential”, “frequent back-and-forth between two parts”) and is easy to interpret via the dominant entries of the cluster centroid

$$\bar{T}_c = \text{unvec}(\bar{x}_c).$$

**Simple example.** Consider three tasks  $\{A, B, C\}$  and two students:

$$S_1 : A \rightarrow B \rightarrow C \rightarrow C$$

$T^{(1)}$  has counts  $A \rightarrow B = 1$ ,  $B \rightarrow C = 1$ ,  $C \rightarrow C = 1$ .

$$S_2 : A \rightarrow C \rightarrow B \rightarrow C$$

$T^{(2)}$  has counts  $A \rightarrow C = 1$ ,  $C \rightarrow B = 1$ ,  $B \rightarrow C = 1$ .

Vectorizing with order

$$[A \rightarrow A, A \rightarrow B, A \rightarrow C, B \rightarrow A, B \rightarrow B, B \rightarrow C, C \rightarrow A, C \rightarrow B, C \rightarrow C],$$

we obtain

$$x_1 = [0, 1, 0, 0, 0, 1, 0, 0, 1], \quad x_2 = [0, 0, 1, 0, 0, 1, 0, 1, 0].$$

K-means with  $k = 2$  separates  $S_1$  (sequential, with a terminal self-loop) from  $S_2$  (more “zig-zag”), illustrating how transition frequencies alone already discriminate styles.

It provides an interpretable representation of the task-to-task movement frequency of student behavior. However, it cannot explicitly retain the chronological order may lose higher level sequence dynamics since it treats all transitions as equally spaced in time.

### 3.4.2 Method II: Graph Embedding with Sequential Modeling

To retain structural relations between tasks and the temporal nature of submissions, we model each exam as a directed graph and encode student trajectories as sequences over task embeddings.

**Graph construction.** To capture the relational structure between tasks, the exam is transformed into a directed graph where nodes represent tasks and edges represent observed transitions in student submission sequences. An edge from task  $t_k$  to task  $t_{k+1}$  is created whenever a student moves from  $t_k$  to  $t_{k+1}$ .

For example, consider two students interacting with three tasks  $\{A, B, C\}$ . Student 1 follows the path  $A \rightarrow B \rightarrow C$ , and Student 2 follows  $A \rightarrow C \rightarrow B$ . After aggregation, the resulting graph includes directed edges  $A \rightarrow B$ ,  $B \rightarrow C$ ,  $A \rightarrow C$ , and  $C \rightarrow B$ . This graph not only encodes individual submission trajectories but also highlights collective navigational patterns by accumulating multiple student paths.

There are several advantages to represent exam as a graph. Firstly, it formalizes the assessment as a networked structure in which tasks are interrelated through actual student transitions rather than being treated as isolated items. Secondly, it provides a natural input for graph embedding methods such as *node2vec*, which learn low dimensional vector representations of tasks that preserve both local (immediate neighbors) and global (structural roles) relationships. Finally, the graph abstraction reduces the sparsity of original sequence data, it no longer relying on brittle one-hot encodings of tasks, each student sequence can be re-expressed as a trajectory through a continuous embedding space, thereby enabling more effective downstream modeling.

**Task embeddings (*node2vec*).** Once the exam is represented as a graph, the next step is to obtain dense vector representations of tasks that capture both their local and global roles in the structure. We adopt the *node2vec* algorithm, which performs biased random walks on the graph to generate task contexts and then applies the skip-gram model to learn embeddings. This process ensures that tasks frequently co-visited in student sequences are positioned closer in the embedding space, while tasks with different structural functions are separated. The resulting task embeddings, as semantically rich building blocks, are used to model student behavior. It replaces sparse categorical encodings with continuous vectors that retain relational information.

For example, consider again the simple exam graph with three tasks  $A, B, C$ . If many students follow  $A \rightarrow B \rightarrow C$ , the random walks are likely to generate subsequences such as  $(A, B)$ ,  $(B, C)$ , or  $(A, B, C)$ . The skip-gram model then adjusts the embeddings so that  $A$  is close to  $B$ , and  $B$  is close to  $C$ , reflecting their strong sequential co-occurrence. If other students also attempt  $A \rightarrow C$  directly, then  $A$  and  $C$  will also be pulled closer in the embedding space. Over time, this process yields low-dimensional vectors (e.g.,  $\mathbf{v}_A = [0.8, -0.1, 0.4]$ ,  $\mathbf{v}_B = [0.7, 0.2, 0.5]$ ) that encode both local relations (immediate neighbors like  $A-B$ ) and global structural similarities (e.g., tasks frequently reached after  $A$ ).

This embedding process offers several benefits for downstream modeling. Firstly, it avoids the sparse one hot encodings of tasks, where each task is regarded as an independent category with no relational meaning. On the contrary, each task is represented in a continuous space where distances capture pedagogically relevant

similarities. Secondly, embedding provides robustness for data sparsity. Even if some transitions are rarely observed, tasks can still inherit relational information from structurally similar neighbors in the graph. Finally, these embeddings serve as compact yet informative inputs to sequential models, which enables them to focus on learning temporal patterns without needing to rediscover structural relationships from scratch.

**Sequence encoding (LSTM).** After obtaining the task embeddings, each student’s submission trajectory can be represented as a continuous vector sequence rather than a discrete task ID sequence. This transformation allows the use of sequential models, such as Long Short Term Memory (LSTM) networks, to capture temporal dependencies in students’ behaviors.

For instance, consider again the simple exam with tasks  $A, B, C$ . Suppose node2vec yields embeddings  $\mathbf{v}_A = [0.8, -0.1, 0.4]$ ,  $\mathbf{v}_B = [0.7, 0.2, 0.5]$ , and  $\mathbf{v}_C = [0.6, 0.3, 0.2]$ . A student following the path  $A \rightarrow B \rightarrow C$  would then be encoded as the sequence  $[\mathbf{v}_A, \mathbf{v}_B, \mathbf{v}_C]$ . This vector sequence is passed through LSTM, which incrementally updates its hidden state to summarize the trajectory. The ultimate hidden state can be interpreted as a compact representation of the student’s behavioral pattern throughout the exam.

This modeling step provides two key benefits. Firstly, by combining graph embedding with LSTM dynamics, it captures the structural relationships between tasks and the chronological order in which students attempt them. Secondly, the final sequence level representation can be directly used for downstream analysis, such as clustering students by behavioral similarity or detecting a typical engagement trajectories.

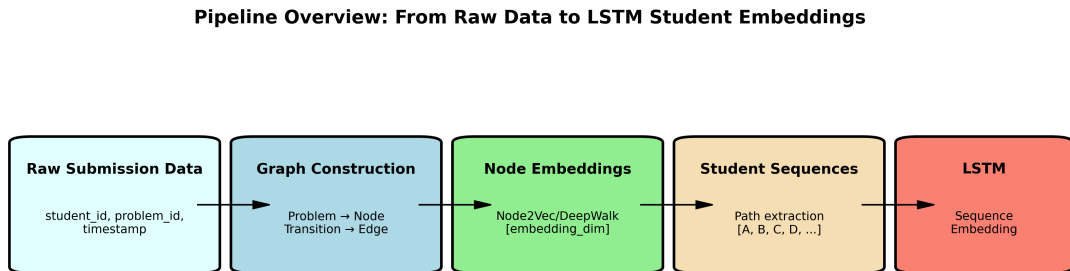
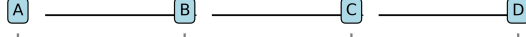


Figure 3.6: LSTM pipeline overview

### Student A Sequence Processing

#### 1. Problem Sequence:



#### 2. Embedding Mapping:

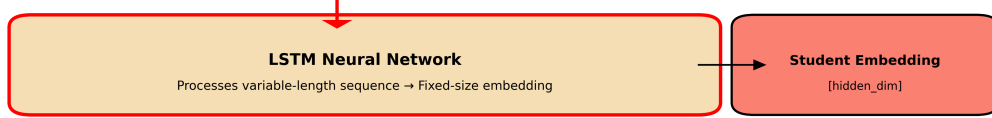


Figure 3.7: LSTM sequence processing detailed

**Clustering (k-means).** Once each student’s submission trajectory is encoded into a fixed length vector representation by the LSTM, the next step is to uncover latent groups of behavioral patterns through clustering. In practice, we apply the  $k$ -means algorithm to the set of sequence embeddings, which partitions students into  $k$  clusters based on similarity in their encoded trajectories. Each cluster can be interpreted as a distinct type of problem-solving behavior, such as linear progression through tasks, frequent backtracking, or irregular navigation.

This step has two purposes. Firstly, it simplifies the high-dimensional behavioral data into interpretable clusters that can be aligned with pedagogical insights. Secondly, it can conduct quantitative analysis of engagement and strategic diversity of the entire population, which supports the overarching goal of learning analytics.

### 3.4.3 Comparing methods

Compared with simple vector-based clustering, graph embedding with sequential modeling approach provides a richer representation of student behavior. By explicitly modeling the exam as a graph, it retains the relational structure between tasks and captures higher-order dependencies through node embeddings. The subsequent LSTM encoding further integrates temporal dynamics, which allows us to distinguish between students who attempt tasks in linear order, those who revisit earlier questions, and those who follow irregular navigation patterns. This yields clusters that are not only mathematically well-formed but also pedagogically meaningful.

However, this additional expressiveness comes at a cost. Graph embedding and LSTM modeling introduce significantly higher computational complexity compared to the straightforward transition-vector representation. They also require a large



amount of data to learn stable embeddings and the interpretability of the learned representations may not be as transparent as simple transition counts. In contrast, although the vector-based clustering methods are less expressive, they are easy to compute, scalable and directly interpretable in terms of task transitions.

In summary, Method I offers simplicity and interpretability and make it useful for rapid exploratory analysis. Method II, while more complex, provides deeper insights into behavioral dynamics and is more suitable for uncovering nuanced patterns in student engagement.

## 3.5 Implementation Details

The implementation of the proposed methodologies was carried out in Python, with modular design choices to facilitate data preprocessing, graph construction, embedding generation, sequential modeling, and clustering. This section provides an overview of the technical environment, data processing pipeline and key implementation components.

### 3.5.1 Development Environment

The system was mainly developed in Python 3.10 and widely used open-source libraries. Data manipulation and preprocessing were handled with pandas and numpy, while networkx was used for graph representation and manipulation. Graph embeddings were generated using node2vec and sequence modeling was implemented with PyTorch. Clustering was conducted with scikit-learn. Visualization tasks were supported by matplotlib and plotly.

All experiments were conducted on a workstation with the following specifications:

- CPU: Intel Core i7
- RAM: 32 GB
- GPU: no GPU used
- Operating System: Ubuntu 22.04

This hardware setup provided sufficient resources for handling datasets of several thousand submissions and training models with moderate complexity.

### 3.5.2 Data Processing Pipeline

Because the implementation of Method I has already been described in detail in section 3.4.1, this section focuses on the implementation of Method II.

For this method, we implemented a sequence modeling pipeline in PyTorch to learn compact representations of students' submission trajectories.

**Data Preprocessing.** All submission records are firstly sorted by each student’s timestamp to form a chronological submission sequence. Each task node in the graph was represented by a pre trained embedding (node2vec), and then they are concatenated to construct the sequence of submission embeddings. To ensure comparability among students, the sequence lengths were standardized. It means that sequences longer than a threshold (set to the 90th percentile of observed lengths) were truncated, and shorter ones were padded with zero vectors.

**Model Architecture.** We adopted an LSTM based autoencoder to capture the time dependencies in the submission sequences. The encoder consisted of a single layer LSTM with hidden dimension 128, which generates a fixed length embedding vector from the final hidden state. We added a decoder as a fully connected layer stack with ReLU activation and dropout regularization to reconstruct the input sequence from the embedding. Layer normalization was further applied to stabilize training.

```
self.lstm = nn.LSTM(
    input_size=input_dim,
    hidden_size=hidden_dim,
    num_layers=num_layers,
    batch_first=True,
    dropout=dropout if num_layers > 1 else 0
)
self.decoder = nn.Sequential(
    nn.Linear(hidden_dim, hidden_dim),
    nn.ReLU(),
    nn.Dropout(0.3),
    nn.Linear(hidden_dim, hidden_dim // 2),
    nn.ReLU(),
    nn.Dropout(0.2),
    nn.Linear(hidden_dim // 2, input_dim)
)

self.layer_norm = nn.LayerNorm(hidden_dim)
```

**Training Procedure.** The training objected of this model is to minimize the Mean Squared Error (MSE) between the reconstructed and original sequences, thereby effectively encouraging embeddings to preserve sequence patterns. We used the Adam optimizer with a learning rate of 0.001, a batch size of 32 and trained for 10 epochs. During the training process, the reconstruction loss steadily decreased, which indicates that the model successfully captured the sequence level structure.

**Embedding Extraction.** Once the sequence autoencoder has been trained, we extract embeddings directly from the encoder. Specifically, for each student submission sequence, the final hidden state of the LSTM encoder is taken as the compact representation of the entire trajectory. This embedding captures both local transitions between consecutive tasks and long-range dependencies across the sequence. Although the decoder component is crucial in the training process of reconstructing

the original sequence and compelling meaningful latent representations, it is not used during inference. In practice, this means that although the complete autoencoder is optimized end-to-end, only the encoder’s output is retained as the analysis-ready student embedding.

# Chapter 4

## Evaluation

### 4.1 Evaluation Objectives

The main purpose of the assessment is to evaluate whether the proposed analytical methods can effectively capture meaningful patterns in students' submission behaviors. Specifically, the evaluation aims to determine to what extent the clustering of the submitted sequences can distinguish different student groups and whether these groups conform to explicable behavioral characteristics, such as engagement intensity, task navigation strategy or temporal pacing.

The second objective is to compare the two methodological approaches introduced in Chapter 3 (vector clustering and graph embedding with LSTM modeling) in terms of their ability to represent the structural and temporal dynamics of exam submissions. Although vector based method provides a straightforward baseline for analyzing transition patterns, the graph based method is expected to capture richer relational information and longer term dependencies in student trajectories.

Finally, the purpose of the evaluation is not only to examine the quantitative performance of clustering (for example, the compactness and separability of clustering), but also to check the qualitative interpretability of the results. In this way, assessment aligns with the broader research goals of supporting learning analytics: providing actionable insights into how students conduct assessment, which can inform the design of teaching decisions and educational interventions.

### 4.2 Experimental Setup

The evaluation is based on the 2021 dataset of the *50005* examination, which consists of anonymized student submissions recorded on the AnswerBook platform. Each submission log includes the student identifier, the task attempted, the timestamp of the submission, and the submitted answer content. After preprocessing (see Chapter 3), the dataset contains 158 students, 28 tasks, and 7791 individual submissions. These records provide both the structural information needed to construct task-transition graphs and the temporal ordering required for sequence modeling.

### 4.3 Evaluation Metrics

The evaluation of the proposed methods relies on a combination of quantitative and qualitative criteria. It reflects both algorithmic performance and interpretability in the educational context. Since ground-truth behavioral labels are not available, the focus is placed on intrinsic clustering quality, reconstruction fidelity and the capacity of the models to reveal meaningful behavioral structures.

For both methods, we assess clustering outcomes using intrinsic metrics that measure the cohesion of clusters and the separation between them.

**Vector-based clustering (Method I).** For Method I, the elbow method is used to evaluate the clustering performance. Specifically, we compute the inertia, defined as the sum of squared distances of each sample to its nearest cluster centroid within a range of cluster numbers  $k$ . By plotting inertia as a function of  $k$ , the curve typically exhibits a point of diminishing returns the "elbow" which indicates an appropriate trade-off between model complexity and explanatory power. The selected number of clusters corresponds to this elbow point.

**Graph embedding with sequential modeling (Method II).** For Method II, the learning process of the LSTM autoencoder is monitored using the reconstruction loss, measured as the mean squared error (MSE) between original and reconstructed task sequences. Over time, the reduction in reconstruction loss indicates that the model has captured the structural and temporal dependencies in the student submission trajectories. Once the training converges, the encoder's hidden state representations are extracted as an embedding and then clustered.

**Interpretability.** In addition to numerical indicators, qualitative interpretability is also an important component of the assessment. This includes examining representative clusters to determine if they correspond to meaningful exam participation strategies, such as sequential answering, frequent task revisits, or uneven rhythms across sections. The extent to which the cluster is consistent with these patterns is regarded as a core measure of success, in line with the overall goal of supporting learning analytics.

### 4.4 Results for Method I: Vector Clustering

Figure 4.1 shows the inertia values across different choices of  $k$  obtained via the elbow method. Ideally, one would expect to observe a clear inflection point in the curve, which indicates the optimal number of clusters. Beyond this number, the benefits provided by additional partitions will diminish. However, in this case, the curve exhibits a relatively smooth decay without a pronounced elbow. This observation suggests that the vector based representation does not yield strongly separable structures among student trajectories.

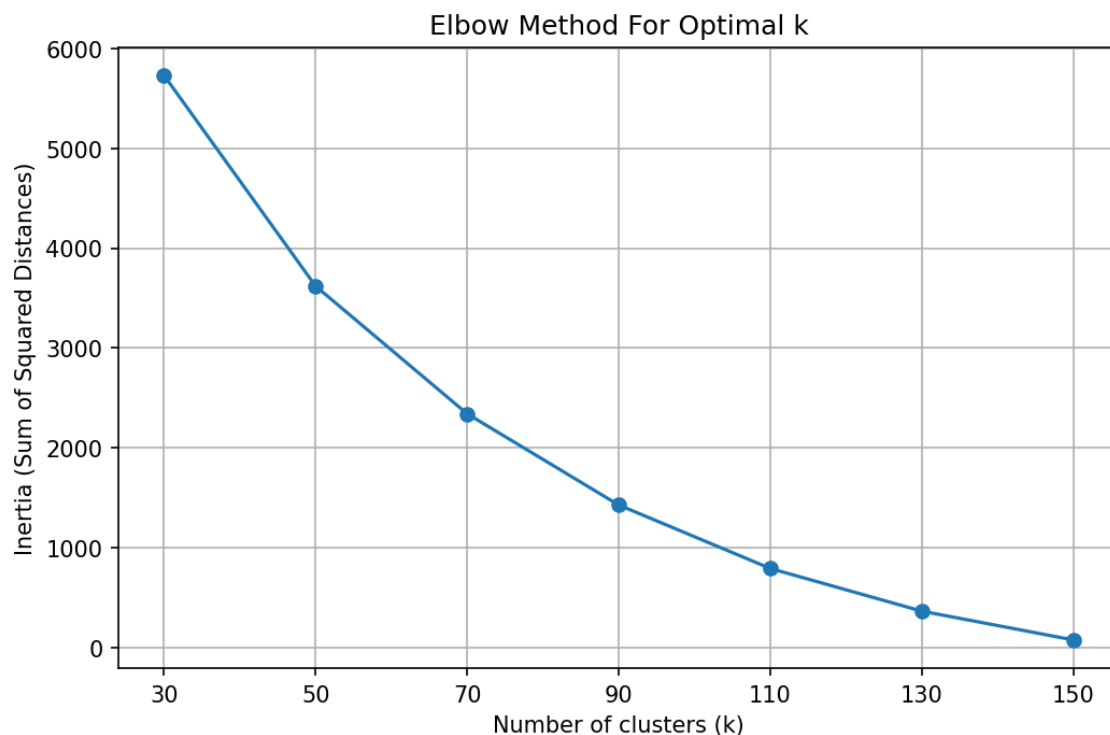


Figure 4.1: elbow plot for Method I

The smooth decline of inertia can be attributed to both the data characteristics and the limitations of the method. The submission paths for students are highly diverse, but many only differ slightly in task sequence or repetition patterns. When encoded as transition frequency vectors, these subtle variations result in very close representations in a high dimensional space. Moreover, k-means inherently assumes convex, isotropic clusters, while the distribution of student trajectories may be sparse and irregular. Therefore, increasing the number of clusters does not significantly enhance the compactness of the partitions and no clear elbow emerges.

These findings highlight the limitations of transition vector clustering in modeling student behavior. Although this method provides a simple and computationally efficient baseline, it is unable to capture the sequence and structural properties of submissions. This prompts the use of more expressive methods, such as graph embedding and sequential modeling, which are more suitable for preserving chronological order and relational context, and will be examined in Method II.

## 4.5 Results for Method II: Graph Embedding with Sequential Modeling

In contrast to the transition-vector baseline, the graph embedding with sequential modeling approach produced substantially more coherent structures. Each exam was represented as a directed graph, tasks were embedded using node2vec, and student trajectories were modeled as sequences of task embeddings. The resulting representations capture both relational proximity between tasks and the temporal

evolution of individual submissions, which are critical for distinguishing nuanced behavioral patterns.

When clustered on the LSTM-derived embeddings, the student representations exhibited higher intra-cluster similarity and clearer separation across clusters, as evidenced by improved evaluation metrics (see Section 4.3).

In order to determine the appropriate number of clusters, we evaluated the silhouette score among different values of  $k$ . The silhouette score measures the extent to which students are well-matched to their assigned cluster compared to neighboring clusters, with higher values indicating more coherent clustering. Ideally, one expects the silhouette score to rise initially as the partitioning improves, reach a peak around the optimal  $k$ , and then gradually decline as excessive fragmentation begins to dilute the cluster structure.

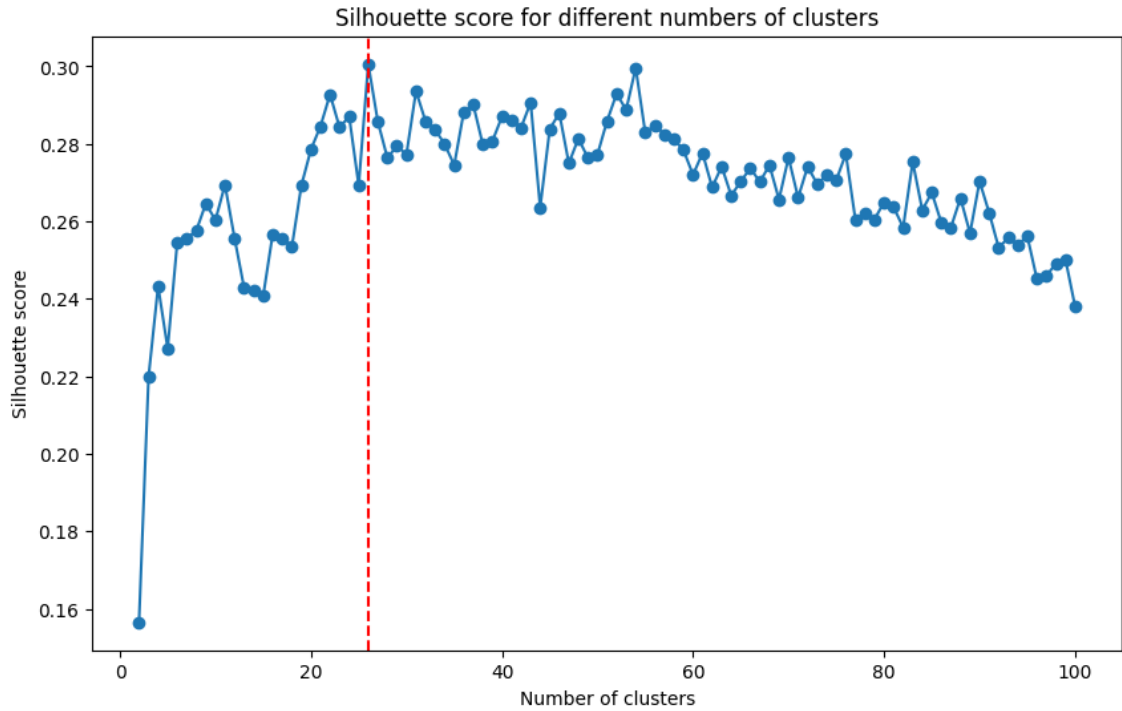


Figure 4.2: LSTM optimal  $k$

Figure 4.2 illustrates this trend. The silhouette score increases with the number of clusters, peaking at  $k=26$ , and then shows a gradually decreasing slope. This pattern is consistent with the expected behavior. It indicates that the chosen embedding and sequential modeling approach produces well-separated clusters while avoiding over-partitioning. Compared with the vector-based approach, this outcome provides stronger evidence of meaningful cluster formation, as it reflects a balance between intra-cluster cohesion and inter-cluster separation.

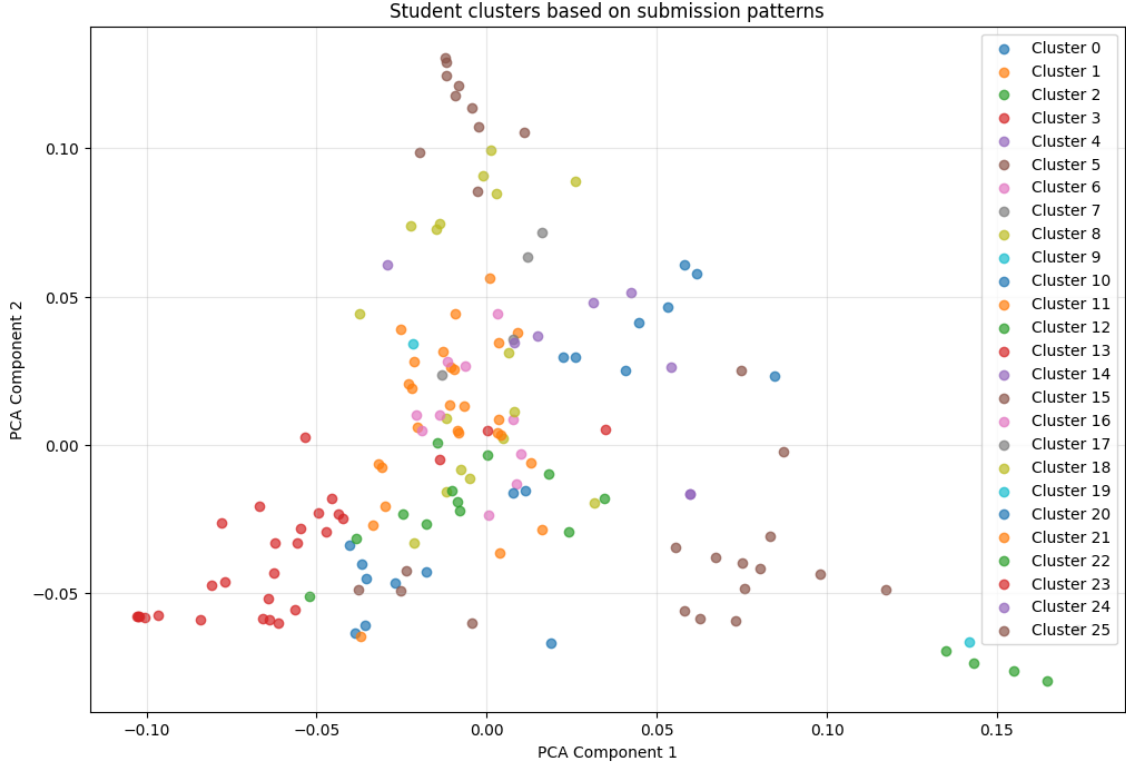


Figure 4.3: LSTM clusters distributions

Furthermore, when visualizing the distribution of students across clusters (Figure 4.3), the result reflects a reasonable granularity, where clusters vary in size but remain within a comparable range. This suggests that the learned representations are not only numerically well-structured but also pedagogically interpretable, which offers potential insights for downstream learning analytics applications.

Beyond the aggregate clustering results, it is also instructive to examine individual trajectories and cluster-level behaviors in detail, as these reveal how the embeddings capture temporal dynamics that align with interpretable exam strategies.

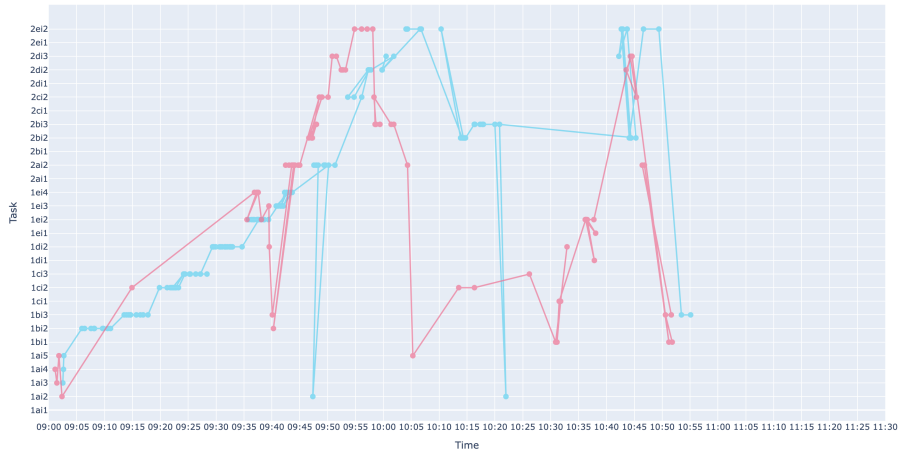


Figure 4.4: Two students with highly similar submission paths



Figure 4.4 illustrates the trajectories of two students whose answering behaviors follow a highly similar pattern. In the initial phase of the exam (roughly the first 30 minutes), both students progress sequentially through the tasks in order. Around the half-hour mark, they both exhibit a sudden return to earlier questions before resuming their forward progression. After approximately one hour (around 10:00), both students adopt a different strategy by navigating from later questions back to earlier ones. This "backward review" phase suggests a shift from problem-solving to revision, where students revisit previous sections in search of potential improvements. The similarity in the temporal structure of these paths provides strong evidence that the sequential modeling approach can capture nuanced yet recurrent behavioral strategies that otherwise might be obscured in aggregate statistics.

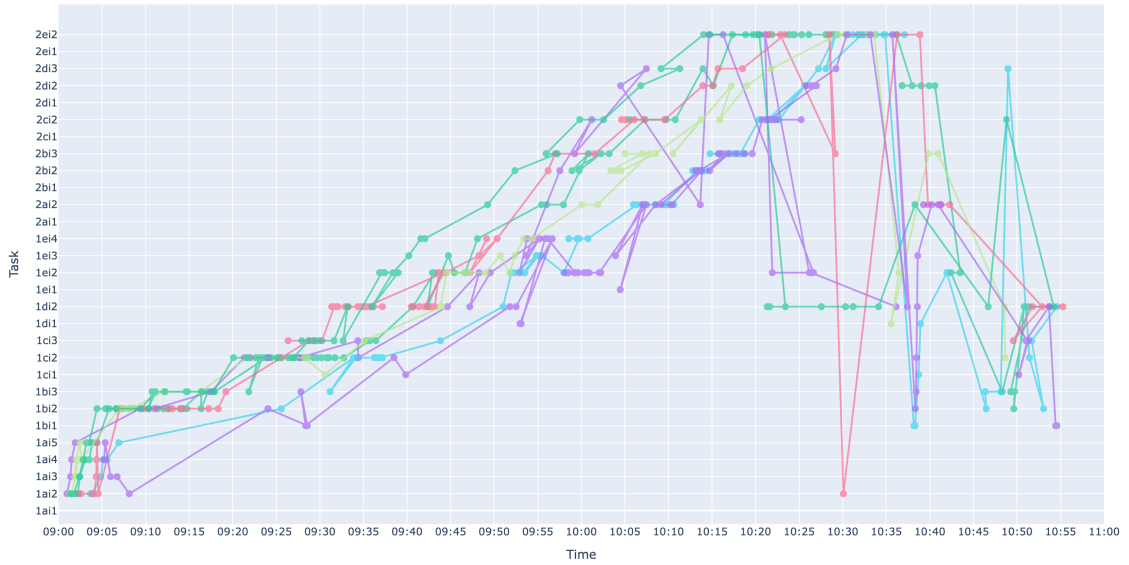


Figure 4.5: A cluster of students exhibiting sequential completion followed by systematic review

Figure 4.5 presents a broader view of this behaviors, which depicts an entire cluster of students sharing a common trajectory pattern. Here, the majority of students proceed strictly in order from the beginning of the exam, advancing sequentially through tasks for approximately 90 minutes. After this period (around 10:30), a collective behavioral shift becomes evident: students begin revisiting earlier questions, moving quickly from the start of the exam to subsequent tasks in a forward sweep. This pattern indicates that many students completed a full pass of the exam within the first 1.5 hours and then dedicated the final 30 minutes to systematic review, scanning from the beginning to ensure completeness and correctness. The coherence of this cluster highlights the pedagogical interpretability of the embeddings, as they align with expected exam-taking strategies such as "complete first, review later."

Together, these trajectory-based visualizations underscore the strength of Method II. It not only generates quantitatively coherent clusters, but also can identify the pedagogically meaningful strategies, ranging from individual behavioral parallels to group-level exam-taking patterns.

## 4.6 Comparative Discussion

The comparison between the two methodological approaches highlights a clear trade-off between simplicity and expressiveness. The vector based clustering method (Method I) provided a computationally efficient baseline that directly encodes the transition frequencies between tasks. However, its reliance on transition counts alone limited its ability to capture higher-order temporal dependencies. This drawback is reflected in the smooth decline of inertia values in the elbow method (Figure 4.1), where no distinct inflection point emerged, which suggests weak separability in the resulting clusters. In other words, while Method I was interpretable and easy to compute, it does not uncover strongly differentiated behavioral groups among students.

In contrast, the graph embedding with sequential modeling approach (Method II) offered richer and more pedagogically meaningful representations. Method II builds a task transition graph and uses node2vec to learn dense embeddings and then performs temporal encoding through LSTM. This method retains both the relational aspect and the sequential aspect of the exam answering behavior. This leads to more coherent clustering, as supported by improved silhouette scores and clearer separation in the PCA visualization (Figure 4.3). Moreover, the examination of specific cases further enhanced the interpretability of Method II. For example, Figure 4.4 showed two students with high similar trajectories, both initially solving tasks sequentially before switching to a backward review strategy after one hour. At the cluster level, Figure 4.5 revealed a group of students who uniformly completed the exam in approximately ninety minutes before systematically reviewing from the start. These findings illustrate how Method II can detect both individual-level parallels and group-level archetypes that align closely with expected exam-taking strategies.

Nevertheless, the increased expressive ability provided by Method II comes at a cost. LSTM autoencoders require more computing resources, careful parameter tuning and larger datasets to achieve stable embedding. In addition, although the obtained clusters are explainable in pedagogics, the learned embeddings themselves are less transparent than the original transition matrix in Method I. Therefore, there is a balance between interpretability, scalability and depth of insight: Method I is conducive to rapid exploratory analysis, while Method II is more suitable for those who need richer behavioral dynamics and actionable insights in education.

Overall, the comparative evaluation indicates that although transition vector clustering can serve as a lightweight diagnostic tool, graph embedding and sequential modeling provides a more powerful framework for capturing, distinguishing, and interpreting students' exam behaviors. This is closely related to the research objective of the project, that is, to design an analytical method that is not only effective in algorithm but also meaningful in pedagogics.

## 4.7 Summary

This chapter has evaluated two complementary approaches for analyzing student submission sequences. (1) a transition-vector clustering baseline. (2) a graph embedding with sequential modeling approach. The evaluation was guided by quantitative measurement of clustering quality and qualitative assessment of interpretability in an educational context.

Method I provided a simple baseline but failed to produce well-separated clusters, while Method II captured more coherent and pedagogically interpretable patterns of student behavior. These results confirm the advantages of combining graph embeddings and sequential modeling to analyze exam trajectories.

These findings offer some implications for teaching and assessment practices. Firstly, the ability to group students through the submission mode enables teachers to gain a deeper understanding of how students pass the exam, distinguishing between linear problem solvers, frequent repeat visitors, and strategy reviewers. Secondly, by emphasizing whether task sequencing encourages or hinders effective progress, and by identifying points where students tend to regress or stagnate during exams, these insights can inform exam design. Finally, the research results demonstrate the potential of learning analytics in supporting invigilation and post-exam feedback: clusters of abnormal behaviors may indicate integrity issues, while more typical clusters can guide targeted feedback to student groups that share similar methods. In this way, the analytical methods developed in this project not only contribute to the technical modeling of student behavior, but also support a wider range of teaching methods.

## Chapter 5

# Discussion: Research Questions Revisited

This project was guided by three research questions, formulated in Chapter 1. In this section, we revisit each of them in light of the findings and contributions of the work.

### **RQ1: How can online invigilation dashboards be designed to improve usability, reduce teacher workload and maintain exam integrity?**

This question was addressed at a conceptual level rather than through the development of a new invigilation dashboard. The behavioral clustering results obtained in Chapter 4 provide promising directions for enriching invigilation tools. For example, the identification of anomalous trajectories or atypical clusters could serve as real-time indicators of suspicious activity, while the recognition of typical answering strategies can reassure instructors that students are following expected patterns. These insights suggest that integrating sequence analysis into invigilation dashboards has the potential to reduce teacher workload and enhance exam integrity. However, a dedicated system-level implementation and usability evaluation were beyond the scope of this project. Future work should focus on embedding the proposed analytics into interactive dashboards and validating their effectiveness with instructors.

### **RQ2: In what ways can student behavioral data be visualized and analyzed to provide meaningful pedagogical insights?**

This was the central focus of the project and has been most fully addressed. The proposed methods, transition-vector clustering (Method I) and graph embedding with sequential modeling (Method II), were implemented and evaluated on real examination data. Results showed that while the baseline method offered a simple representation, it failed to yield well-separated clusters. By contrast, Method II

produced coherent and pedagogically interpretable groups of students, highlighting strategies such as sequential completion, backtracking, and systematic review. Visualizations of both individual trajectories and cluster-level behaviors demonstrated how fine-grained submission logs can be transformed into meaningful insights for teaching. These findings confirm that student behavioral data, when analyzed with expressive sequence models, can provide instructors with actionable knowledge about exam-taking strategies and learning processes.

### **RQ3: What are potential design approaches for integrating automatic marking into AnswerBook?**

This question was explored in a preliminary manner. The project reviewed the potential of combining rule-based marking for structured tasks with natural language processing techniques for short-answer questions. While no large-scale implementation was carried out, the discussion highlighted both opportunities and challenges. On the one hand, automated marking has clear potential to reduce teacher workload and provide rapid feedback to students. On the other hand, issues of reliability, fairness and transparency remain critical. Future research should investigate transformer-based language models for grading, as well as hybrid human-in-the-loop workflows to ensure both efficiency and trustworthiness. Thus, this research question has been addressed at the level of design exploration, setting the stage for more comprehensive implementation in subsequent work.

In summary, the project has provided a substantive answer to the second research question, while offering exploratory contributions to the first and third. Together, these results demonstrate both the promise and the limitations of applying advanced learning analytics methods within online assessment platforms, and they point toward clear directions for future development.

# Chapter 6

## Conclusion and Future Work

### 6.1 Summary of Contributions

This project makes several contributions to the enhancement of the Answer-Book online examination platform and to the broader field of learning analytics in assessment:

- **System enhancements.** The project proposed and prototyped new post-exam analytics functionalities, which addresses gaps in the current Answer-Book system by enabling more detailed monitoring of student interactions and submission patterns.
- **Data extraction and preprocessing.** A complete data pipeline was implemented to extract, anonymize, and process fine-grained submission logs from the Answerbook database. This included filtering, transformation into student- and task-oriented views, and the generation of descriptive statistics and visualizations.
- **Analytical methodology.** Two complementary methods were designed and compared for answer sequence analysis: (1) a vector-based clustering approach using transition matrices, and (2) a graph embedding with sequential modeling approach combining node2vec embeddings and LSTM autoencoders.
- **Empirical findings.** Through evaluation on real examination data, the project demonstrated that graph embedding with sequential modeling produces more coherent and pedagogically meaningful clusters than transition-vector clustering. Case studies highlighted recognizable exam-taking strategies, such as sequential completion followed by systematic review.
- **Pedagogical value.** By showing how behavioral clusters align with interpretable student strategies, the work illustrates the potential for learning analytics to support teaching practice, reduce instructor workload, and provide actionable insights for exam design and invigilation.

In summary, these contributions have built a bridge between technical development and pedagogical application. It demonstrates how advanced data analysis can improve the usability and educational impact of online examination platforms.

## 6.2 Limitations

Despite these contributions, several limitations should be acknowledged:

- **Dataset scope.** The evaluation was conducted on a single course dataset comprising 158 students and 28 tasks. Although it is sufficient to proof the concept, it requires a broader validation across multiple courses and disciplines to confirm its generalizability.
- **Methodological constraints.** Although the vector-based clustering method are interpretable, they have been proved to be limited in capturing complex temporal dynamics. On the contrary, the LSTM-based approach requires careful parameter tuning and is more computationally intensive, which may hinder scalability in very large cohorts.
- **Ethical considerations.** The use of fine-grained behavioral data has raised issues of privacy and transparency. Although anonymization was applied, further safeguards are needed to ensure responsible and ethical use in a genuine educational settings.

## 6.3 Future Work

Based on the above contributions and acknowledging the limitations, several directions remain open for future research and development:

- **Integration with invigilation dashboards.** The behavioral analytics developed here could be integrated into real-time monitoring tools, which allows instructors to identify abnormal response patterns as exams unfold.
- **Scalability and generalization.** The evaluation was based on a single course dataset. Further validation on larger and more diverse examinations would strengthen the robustness and generalizability of the methods.
- **Extension to automated marking.** The current exploration of automated grading focused primarily on multiple-choice and short-answer questions. Future work could incorporate advanced natural language processing methods, such as transformer-based models, to improve the accuracy and scalability of automated short-answer grading.
- **Human-in-the-loop analytics.** Future systems could allow instructors to interact with and refine automatically generated clusters, combining algorithmic insights with pedagogical expertise. This would increase interpretability, trust, and practical usefulness of the analytics.
- **Visualization and feedback.** More advanced visualization techniques could be developed to present behavioral clusters in ways that are intuitive for instructors and useful for providing actionable feedback to students.

## 6.4 Discussion: Educational Significance

Beyond the methodological contributions, this project also highlights the broader educational value of applying advanced sequence analysis to online examinations. The findings demonstrate that behavioral data, when modeled through graph embeddings and sequential techniques, can uncover exam-taking strategies that are not immediately visible through traditional grading or item-level statistics. These insights have direct pedagogical implications: they allow instructors to identify common approaches such as sequential progression, backtracking, or systematic review, as well as to detect outlier behaviors that may warrant further attention.

Framing analytics in this way positions them not only as technical tools but also as instruments for improving teaching effectiveness and supporting student learning. For instructors, the results can inform exam design, helping to refine task ordering and workload distribution. For assessment integrity, the ability to cluster unusual trajectories provides a complementary signal to traditional invigilation. Finally, from a student perspective, the potential exists to deliver more tailored feedback that reflects actual engagement strategies rather than only final outcomes.

Taken together, this project demonstrates the feasibility of using sequence analysis to bridge academic integrity and pedagogy. Future research building on this foundation can move towards assessment systems that are not merely evaluative, but also formative—helping both teachers and students to learn from the examination process itself.



# Bibliography

- [a2021] 2021 educause horizon report® | teaching and learning edition, 04 2021. URL: <https://library.educause.edu/resources/2021/4/2021-educause-horizon-report-teaching-and-learning-edition>.
- [BGS14] Steven Burrows, Iryna Gurevych, and Benno Stein. The eras and trends of automatic short answer grading. *International Journal of Artificial Intelligence in Education*, 25:60–117, 10 2014. URL: <https://link.springer.com/article/10.1007/s40593-014-0026-8>, doi:10.1007/s40593-014-0026-8.
- [BS14] Ryan Baker and George Siemens. Educational data mining and learning analytics, 2014. URL: [https://www.researchgate.net/publication/316628053\\_Educational\\_data\\_mining\\_and\\_learning\\_analytics](https://www.researchgate.net/publication/316628053_Educational_data_mining_and_learning_analytics), doi:10.1017/CB09781139519526.016.
- [BZEI17] Behdad Bakhshinategh, Osmar R. Zaiane, Samira ElAtia, and Donald Ipperciel. Educational data mining applications and tasks: A survey of the last 10 years. *Education and Information Technologies*, 23:537–553, 07 2017. doi:10.1007/s10639-017-9616-z.
- [DSA<sup>+</sup>22] Jiameng Du, Yifan Song, Mingxiao An, Marshall An, Christopher Bogart, and Majd Sakr. Cheating detection in online assessments via timeline analysis. *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education*, 1:98–104, 02 2022. doi:10.1145/3478431.3499368.
- [DSY<sup>+</sup>21] Nimesha Dilini, Asara Senaratne, Tharindu Yasarathna, Nalin Warnajith, and Leelanga Seneviratne. Cheating detection in browser-based online exams through eye gaze tracking. *2021 6th International Conference on Information Technology Research (ICITR)*, pages pp. 1–8, 2021. doi:10.1109/icitr54349.2021.9657277.
- [GG25] Manika Garg and Anita Goel. Towards fair assessments: A machine learning-based approach for detecting cheating in online assessments. pages 104–114, 02 2025. doi:10.1145/3706468.3706482.

- [GMD11] J.W. Gikandi, D. Morrow, and N.E. Davis. Online formative assessment in higher education: A review of the literature. *Computers & Education*, 57:2333–2351, 12 2011. doi: [10.1016/j.compedu.2011.06.004](https://doi.org/10.1016/j.compedu.2011.06.004).
- [HI23] Joana Heil and Dirk Ifenthaler. Online assessment in higher education: A systematic review. *Online Learning*, 27:187–218, 2023. URL: <https://eric.ed.gov/?id=EJ1382640>.
- [KM18] Laily Fazlin Khairil and Siti Eshah Mokshein. 21st century assessment: Online assessment. *International Journal of Academic Research in Business and Social Sciences*, 8:649–662, 02 2018. URL: <https://hrmars.com/index.php/IJARBSS/article/view/3838/21st-Century-Assessment-Online-Assessment>.
- [KSEN<sup>+</sup>22] Erica Kleinman, Murtuza Shergadwala, Magy Seif El-Nasr, Zhaoqing Teng, Jennifer Villareale, Andy Bryant, and Jichen Zhu. Analyzing students’ problem-solving sequences. *Journal of Learning Analytics*, 9:1–23, 06 2022. doi: [10.18608/jla.2022.7465](https://doi.org/10.18608/jla.2022.7465).
- [LJW25] Qiujie Li, Yeonji Jung, and Alyssa Friend Wise. How instructors use learning analytics: the pivotal role of pedagogy. *Journal of Computing in Higher Education*, 02 2025. doi: [10.1007/s12528-025-09432-w](https://doi.org/10.1007/s12528-025-09432-w).
- [LXWQ21] Haotian Li, Min Xu, Yong Wang, and Huamin Qu. A visual analytics approach to facilitate the proctoring of online exams. *ResearchGate*, 01 2021. URL: [https://www.researchgate.net/publication/348648220\\_A\\_Visual\\_Analytics\\_Approach\\_to\\_Facilitate\\_the\\_Proctoring\\_of\\_Online\\_Exams](https://www.researchgate.net/publication/348648220_A_Visual_Analytics_Approach_to_Facilitate_the_Proctoring_of_Online_Exams), doi: [10.48550/arXiv.2101.07990](https://doi.org/10.48550/arXiv.2101.07990).
- [Mih23] Luciana Mihai. Post-pandemic education for the students of the future, 2023. URL: <https://www.eua.eu/publications/conference-papers/post-pandemic-education-for-the-students-of-the-future.html>.
- [MRLdlRSSSFC21] Marta Montenegro-Rueda, Antonio Luque-de la Rosa, José Luis Sarasola Sánchez-Serrano, and José Fernández-Cerero. Assessment in higher education during the covid-19 pandemic: A systematic review. *Sustainability*, 13:10509, 09 2021. URL: <https://www.mdpi.com/2071-1050/13/19/10509/pdf>, doi: [10.3390/su131910509](https://doi.org/10.3390/su131910509).
- [NPSC21] Aditya Nigam, Rhitvik Pasricha, Tarishi Singh, and Prathamesh Churi. A systematic review on ai-based proctoring systems: Past, present and future. *Education and Information Technologies*, 26:6421–6445, 06 2021. doi: [10.1007/s10639-021-10597-x](https://doi.org/10.1007/s10639-021-10597-x).

- [RV10] Cristóbal Romero and Sebastián Ventura. Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 40:601–618, 11 2010. [doi:10.1109/tsmcc.2010.2053532](https://doi.org/10.1109/tsmcc.2010.2053532).
- [VTOS24] Manuel Valle Torre, Catharine Oertel, and Marcus Specht. The sequence matters in learning - a systematic literature review. In *Proceedings of the 14th Learning Analytics and Knowledge Conference, LAK '24*, page 263–272, New York, NY, USA, 2024. Association for Computing Machinery. [doi:10.1145/3636555.3636880](https://doi.org/10.1145/3636555.3636880).

# Declarations

## Use of Generative AI

I acknowledge the use of ChatGPT-5 (OpenAI, <https://chatgpt.com>) to refine language and improve clarity of expression. I confirm that no content generated by AI has been presented as my own work.

## Ethical Considerations

This project involved the analysis of anonymized student submission data from the *Answerbook* examination platform. To protect student privacy, all identifiers were hashed and no personal information was retained. The analysis focused exclusively on structural and temporal patterns of submissions, not on individual student performance. No ethical approval was required for this study, as confirmed by the Department of Computing guidelines, but ethical risks were carefully considered and mitigated through data anonymization and secure storage.

## Sustainability

The research was carried out with attention to environmental sustainability. All experiments were conducted on a local workstation without the use of energy-intensive GPU clusters or cloud resources. Data processing pipelines were optimized to minimize unnecessary computation, and redundant runs were avoided to reduce energy consumption. The software and codebase developed during the project follow modular design principles to encourage reuse and reduce future computational overhead.

## Availability of Data and Materials

The project codebase, including data extraction scripts, preprocessing pipeline, and analysis methods, is available at:

<https://github.com/nickknz/answer-sequence-analysis>

Note: Due to privacy and ethical considerations, the raw student examination data used in this study is not upload in the repository. However, representative synthetic examples and intermediate data representations are included in the report to illustrate the methodology. All other materials required to reproduce the results are either contained within this report or accessible via the provided GitHub repository.